PHARMACOLOGY

Female Sex Hormones & Contraception

SGL 1
Dr. Chro Yaseen Qader
GUE Block

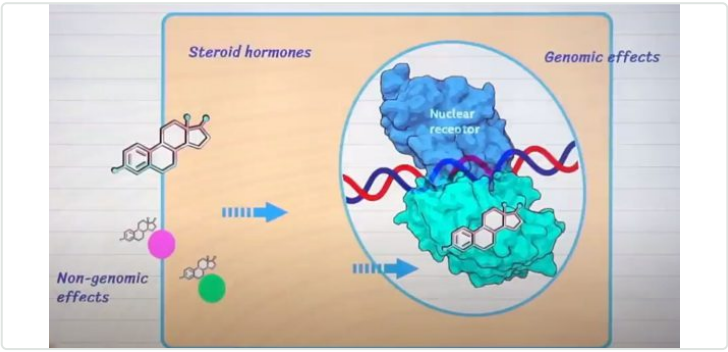
Pharmacology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

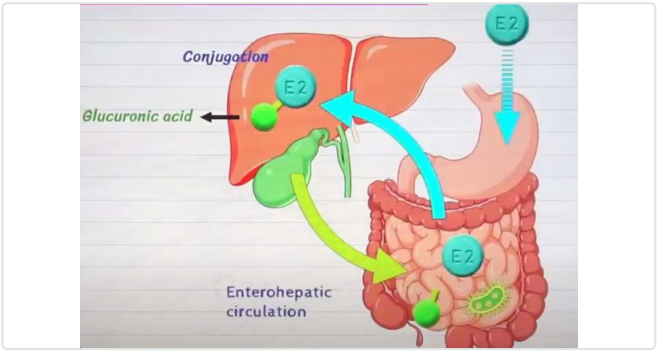
1. Tamoxifen protects which tissue but harms it in Raloxifene's place, and vice versa — bone, breast, endometrium?
2. How does progestin alone (the "mini-pill") prevent pregnancy if it doesn't reliably block ovulation?
3. Within what window must levonorgestrel vs ulipristal emergency contraception be taken?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Hormonal control & mechanism



Steroid hormones act on nuclear receptors (genomic, slow) and some membrane receptors incl. a G-protein-coupled oestrogen receptor (non-genomic, fast).

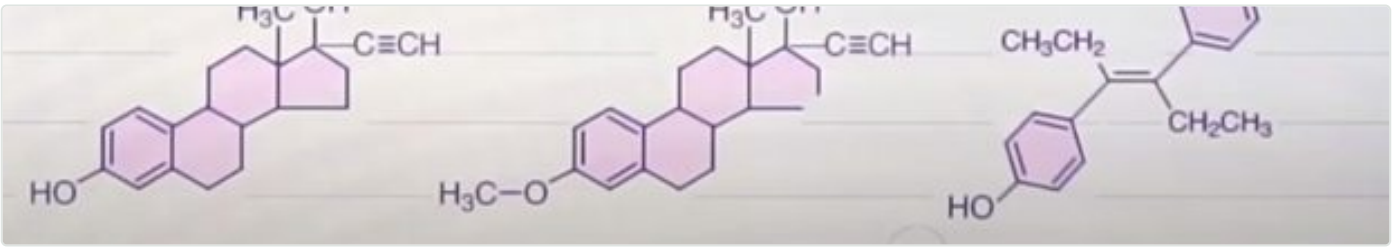


Oestrogens undergo hepatic glucuronide conjugation + **enterohepatic cycling** — bound to albumin and sex-steroid-binding globulin.

- **Oestrogen induces** progesterone-receptor synthesis in uterus, vagina, anterior pituitary, hypothalamus; **progesterone does the opposite** — down-regulates oestrogen receptors in the reproductive tract.
- Negative feedback: oestrogen + progesterone suppress hypothalamic GnRH → ↓LH/FSH → no ovulation (the basis of every hormonal contraceptive).

2 Oestrogens

Class	Examples
Natural	Oestradiol, oestrone, estriol
Synthetic	**Ethinylestradiol** (most common in COCs), mestranol, diethylstilbestrol
Conjugated	Mixture of several oestrogens (used in HRT)



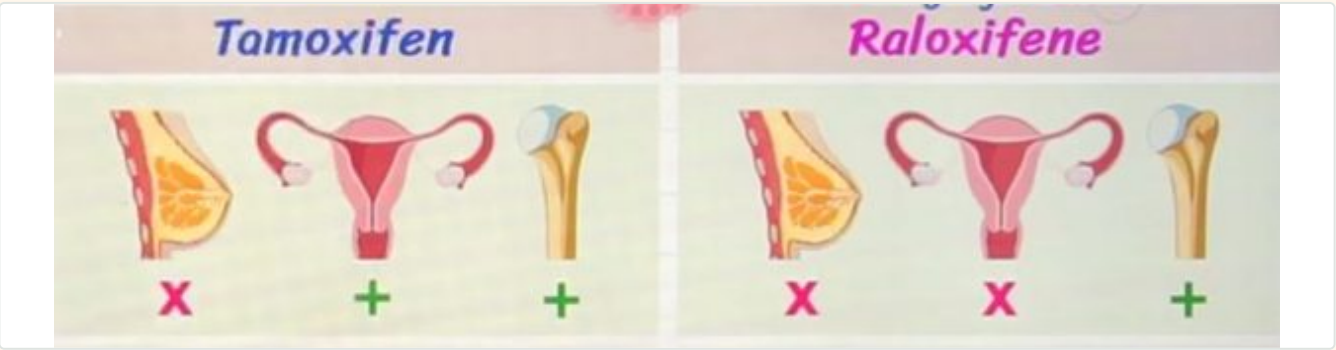
Ethinylestradiol / mestranol / diethylstilbestrol structures

Clinical uses	Adverse effects	Notes
COC (with progestin); dysfunctional uterine bleeding; ovarian failure / delayed puberty; premature menopause; postmenopausal HRT (atrophic vaginitis, hot flush, osteoporosis)	Breast tenderness, nausea/vomiting/anorexia, salt & water retention (oedema), **↑thromboembolism risk**; **endometrial hyperplasia** if unopposed in a woman with a uterus; feminisation if given to males; genital abnormality in offspring if given in pregnancy	**Women with uterus intact need added progestogen** with HRT to prevent endometrial hyperplasia/cancer; women with uterus removed can take oestrogen alone

3 Oestrogen antagonists — comparison grid

Drug	Mechanism	Clinical use
Tamoxifen	SERM — competitive partial agonist, tissue-dependent	ER+ breast cancer (adjuvant post-mastectomy/radiation); chemoprevention in high-risk women
Raloxifene	SERM — competitive partial agonist, tissue-dependent	Postmenopausal osteoporosis; breast-cancer chemoprevention; dysfunctional uterine bleeding
Clomiphene	Pure oestrogen-receptor **blocker** at hypothalamus/pituitary	Covered in full under §5 (ovulation induction)
Letrozole, anastrozole	Aromatase inhibitors — block androgen→oestrogen conversion	ER+ breast cancer (1st/2nd line in postmenopausal women)

DISCRIMINATOR — TAMOXIFEN vs RALOXIFENE (the exact opposite tissue pattern)



	Breast	Uterus/endometrium	Bone
Tamoxifen	**Anti-oestrogenic** (✗ good)	**Oestrogenic** → ↑endometrial cancer risk	Oestrogenic (protective)
Raloxifene	**Anti-oestrogenic** (✗ good)	**Anti-oestrogenic** (safe)	Oestrogenic (protective)

Both ↑ clotting factors / thrombosis risk. Only tamoxifen raises endometrial cancer risk — that's the one fact examiners hang a vignette on.

4 Progestogens

- Natural = **progesterone** (corpus luteum, 2nd half of cycle; placenta in pregnancy); synthetic = **progestin**.
- Acts on nuclear receptors; receptor density is itself controlled by oestrogen.

Generation	Agents	Key feature
1st / 2nd (older)	Norethisterone, norgestrel, levonorgestrel	Testosterone derivatives — **androgenic** activity; metabolised to oestrogenic products
3rd (newer)	Desogestrel, gestodene, norgestimate	↓Androgenic/metabolic effects; used when 1st/2nd-gen cause acne, depression, breakthrough bleeding
4th	**Drospirenone** (aldosterone analogue), cyproterone, dienogest	**Anti-androgenic** — used for acne, hirsutism; drospirenone also anti-mineralocorticoid

Clinical uses	Adverse effects
Contraception (all forms below); HRT add-on to prevent endometrial hyperplasia; endometriosis; endometrial carcinoma; menstrual disorders; threatened abortion/preterm labour (**natural progesterone, dydrogesterone** only)	Acne, fluid retention, weight change, depression, ↓libido, breast discomfort, breakthrough bleeding; ↑thromboembolism

Mifepristone (antiprogestogen) — partial agonist at progesterone receptors, sensitises uterus to prostaglandins; oral; combined with **misoprostol** for medical termination of pregnancy. ADRs: abdominal pain, uterine bleeding, incomplete abortion.

5 Hormonal contraceptives — comparison grid

Method	Composition / regimen	Key point

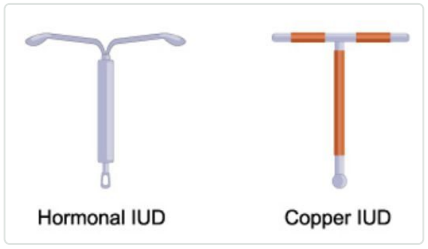
Combined oral (COC)	Ethinylestradiol + progestin; 21–24 active days + 4–7 placebo days (28-day pack); monophasic (constant dose) or triphasic (mimics natural cycle)	Withdrawal bleed occurs in placebo week
Transdermal patch	Oestrogen + progestin	1 patch weekly × 3 weeks, none in week 4
Vaginal ring	Ethinyl oestrogen + progestin	—
Progestin-only pill (POP)	Progestin only, continuous daily ("mini-pill")	Less effective than combined; for breastfeeding, oestrogen-intolerant, smokers
Injectable	**Medroxyprogesterone acetate**, IM/SC every 3 months	Weight gain; delayed return of fertility; osteoporosis/fracture risk
Implant	Etonogestrel subdermal	~3 years' coverage
Progestin IUD	Intrauterine system	3–5 years; avoid in PID / prior ectopic pregnancy
Emergency ("morning-after")	**Levonorgestrel** (preferred) or ethinylestradiol + levonorgestrel; alt. **ulipristal** (progesterone-receptor modulator)	Levonorgestrel: ASAP, within **72h**. Ulipristal: within **5 days (120h)**



Monophasic COC pack



Transdermal contraceptive patch



Hormonal vs copper IUD

6 Mechanism, adverse effects & contraindications of combined contraception

- **Oestrogen** → negative feedback on LH/FSH → blocks ovulation.
- **Progestin** → thickens cervical mucus (blocks sperm transport); withdrawal in placebo week triggers menstrual bleeding.

Common ADRs	Rare but serious (esp. >35y)
Breast fullness, fluid retention, headache, nausea, ↑BP (oestrogen); depression, hirsutism, acne (progestin); ↑cervical cancer incidence reported	Thromboembolism, MI/stroke, cholecystitis/gallstones

Contraindications
Hypertension/IHD · history of embolism/thrombosis/cerebral haemorrhage · breast cancer / oestrogen-dependent neoplasm history · migraine · chronic liver disease/gallstones · diabetes mellitus · obesity/smoking · pregnancy · depression

CAUSES OF PILL FAILURE

Enzyme inducers (rifampin, phenytoin) · broad-spectrum antibiotics (e.g. tetracyclines) · paraffin oil (laxative) — ↓ intestinal absorption

P Must-remember pearls

- **Tamoxifen** is anti-oestrogenic on breast but pro-oestrogenic on endometrium (↑endometrial cancer) and bone; **raloxifene** is anti-oestrogenic on BOTH breast and uterus, pro-oestrogenic only on bone.
- A woman with an **intact uterus** on HRT must get oestrogen + progestogen (prevents endometrial hyperplasia/cancer); a woman with the uterus **removed** can take oestrogen alone.
- Emergency contraception: **levonorgestrel within 72h** (preferred); **ulipristal within 5 days** if later.
- Progestin alone doesn't reliably stop ovulation — its contraceptive effect leans on thickening cervical mucus, which is why POPs are less effective than combined pills.
- Enzyme inducers, broad-spectrum antibiotics, and paraffin oil are the three classic causes of OCP failure.

Q Test yourself

1. A 42-year-old woman on tamoxifen for ER-positive breast cancer presents with postmenopausal bleeding. Endometrial biopsy shows hyperplasia. Which property of tamoxifen explains this?

A. Oestrogenic (agonist) action on the endometrium
B. Suppression of LH/FSH secretion
C. Anti-oestrogenic action on the endometrium
D. Progesterone-receptor blockade
E. Inhibition of aromatase

2. A woman who had a hysterectomy 2 years ago is started on hormone replacement for hot flushes and vaginal dryness. What is the most appropriate regimen?

A. Oestrogen + progestogen
B. Oestrogen alone
C. Progestogen alone
D. Clomiphene
E. Tamoxifen

3. A 26-year-old had unprotected intercourse 4 days ago and wants emergency contraception. Which agent is still appropriate at this point, and levonorgestrel is not?

A. Letrozole
B. Mifepristone
C. Medroxyprogesterone acetate
D. Ulipristal
E. Ethinylestradiol alone

4. A woman on combined oral contraceptives is started on rifampin for TB treatment and becomes pregnant despite good pill compliance. What is the mechanism?

A. Rifampin antagonises the progestin receptor
B. Rifampin displaces the pill from plasma protein binding
C. Rifampin increases sex-hormone-binding globulin
D. Rifampin induces hepatic enzymes that increase contraceptive metabolism
E. Rifampin blocks intestinal absorption of the pill

5. A postmenopausal woman with osteoporosis is being considered for raloxifene instead of tamoxifen for breast-cancer risk reduction. What is the key safety advantage of raloxifene in this context?

A. It does not increase thromboembolism risk at all
B. It does not stimulate the endometrium (no ↑endometrial cancer risk)
C. It has no effect on bone density
D. It cannot be used for osteoporosis
E. It lowers LDL more than tamoxifen

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — C — Tamoxifen is anti-oestrogenic on breast tissue but acts as an oestrogen agonist on the endometrium, driving hyperplasia and raising endometrial cancer risk — the single most tested tamoxifen fact.

2 — B — No uterus means no endometrial hyperplasia risk, so unopposed oestrogen is appropriate and avoids unnecessary progestogen side effects.

3 — B — Ulipristal is effective up to 5 days (120h) post-coitus; levonorgestrel's window is 72h.

4 — B — Rifampin is a classic hepatic enzyme inducer; it accelerates oestrogen/progestin metabolism, lowering contraceptive efficacy — the exact mechanism tested under "causes of pill failure."

5 — A — Raloxifene is anti-oestrogenic on both breast AND uterus (unlike tamoxifen), so it protects bone without raising endometrial cancer risk.

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam